Yuri Gelsleichter

Researcher in Soil and Geomatics

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Nationality: Brazilian, Luxembourgish

Soil and Geomatics Researcher with a Ph.D. in Science, Technology, and Innovation in Agriculture. Expert in **Digital Soil Mapping**, AI-driven **Machine Learning**, Remote Sensing and **Geographic Information System** (GIS) with R, Python, Quantum GIS and Google Earth Engine (JavaScript). Published 14+ peer reviewed articles (148 citations October 2026, h-index 7) and contributed to Open Science projects like Soils4Liberia, OSIRIS and AUSO. Skilled in Open source GIS (QGIS, GDAL, R packages like terra, SF, RSAGA), data visualization, and computational reproducibility. International collaboration across the Americas, Europe, Africa, and Oceania.

Skills

Programming	**R programing**, **Javascript-Google Earth Engine-GEE**, **Bash-Linux**, **Git**, Artificial Intelligence, Big Data, Cloud Computing, Python, SQL
HPC experience	NCAR-Derecho, emory-Saikawa, NERSC
GIS Software	**GDAL**, QGIS, SAGA-GIS, GRASS
Data Science	Soil Spectral Analysis, Applied AI-driven machine learning to soil property prediction, ETL tools: [extraction, transform, and load data], Data Management, Data Visualization, Multivariate Analyses
Digital Skills	Digital Soil Mapping, Hyperspectral Soil Mapping, R Programming, Geographic Information System-GIS, Soil Spectroscopy, Machine Learning with Python and R, Remote Sensing, Soil Classification
Interdisciplinary	Environmental Science, Agricultural Innovation, Precision Agriculture, Science Reproducibility, Methodology Development
Soft Skills	Team Collaboration, Problem-Solving, Facilitated international collaborations and communication as part of the Soil Judging committee in The International Union of Soil Sciences (IUSS)

Experience

Current – Feb 2021 | **Research, post-doc | Hungarian University of Agriculture and Life Sciences (MATE), Gödöllő, Hungary**
- Researching soil classification, mapping, and spectral characterization.
- Taught 60+ graduate students in R programming courses and 40+ in soil amelioration, and protection.
- Advising two doctoral students in Digital Soil Mapping and two in Soil Spectroscopy.
- International collaboration: USA, Brazil, Australia, Hungary, Namibia, Kenya, Tanzania, The Netherlands.

R | Javascript-Google Earth Engine | Teaching

Feb 2021 – Aug 2020 | **Data Scientist | Just.bi, São Paulo, Brazil**
- Designed sampling data for analytical models.
- Implemented Machine Learning models for data analysis and time series.
- Developed Recency, Frequency, Value (RFV) models to enhance customer insights.

Jan 2019 – Aug 2019 | **Research Guest | ISRIC, Wageningen, Netherlands**
- Developed land cover change scenarios for predicting carbon stock in Argentina.
- Mapped large areas using advanced GIS and remote sensing techniques.
- Collaborated with international teams on soil data projects.

Jun 2018 – Sep 2017 | **Graduate Exchange Student | Szent István University (now MATE), Gödöllő, Hungary**
- Contributed to the Universal Soil Classification System.
- Created taxa centroids to compare WRB and Brazilian Soil Classification systems.
- Engaged in the Stipendium Hungaricum Program for international study.

Education

Mar 2016 – Mar 2020: Ph.D. in Science, Technology, and Innovation in Agriculture (Federal Rural University of Rio De Janeiro (UFRRJ), Seropédica, Brazil)
- Specialized in soil classification, mapping, and spectral characterization.
- Developed expertise in R programming, machine learning, and data analytics.
- Engaged in research projects enhancing computational reproducibility.

Aug 2009 – Aug 2015: B.Sc. in Environmental and Sanitary Engineering (University of Southern Santa Catarina (UNISUL), Palhoça, Brazil)
- Acquired foundational knowledge in environmental engineering principles.
- Participated in projects on environmental sustainability and sanitation.
- Completed coursework emphasizing geomatics and coding applications.

Awards, Grants & Scholarships

Research Excellence Programme — Hungarian University of Agriculture and Life Sciences: Soil Research (2026)

Geospatial Hackaton — OpenGeoHub: EO Summer School – Spacetime prediction of Soil Organic Carbon (2025)

Fellowship — CLIFF-GRADS: Research Guest ISRIC (2019)

Scholarship — Stipendium Hungaricum: PhD exchange MATE University (2017-2018)

Scholarship — Science Without Borders: Brazilian Government (International Mobility Program) Szent István University (2013-2014)

Publications

All publications in Google scholar

Selected publications:

National-Scale Multi-Depth Digital Soil Mapping in Namibia Based on Legacy Data 2026
Marina Elda Coetzee, Yuri Andrei Gelsleichter, Ádám Csorba, Erika Michéli
DOI: 10.3390/soilsystems10100111
Country-wide, multi-depth digital soil mapping of Namibia built from legacy soil data, delivering national baseline maps of key soil properties – *Soil Systems*.

Exploratory Insights into Barriers and Open Practices for Computational Reproducibility in Scientific Research 2026
Gelsleichter, Y. A., Banzi, R., Naudet, F., Vinatier, C., Kertész, I., Varga, M.
DOI: 10.12688/f1000research.172013.3
Multidisciplinary survey identifying the major barriers to computational reproducibility and the open practices researchers actually adopt – *F1000Research* (latest indexed version).

High resolution mapping of soil organic carbon in northern Nigerian agricultural lands using comparative machine learning with uncertainty analysis 2026
Mustapha Abdulkadir, Ádám Csorba, Márta Fuchs, Tamás Szegi, Babangida Hammani, Erika Michéli, Yuri Andrei Gelsleichter
DOI: 10.1007/s43621-026-03407-2
Comparison of machine learning algorithms for high-resolution soil organic carbon mapping in northern Nigerian croplands, including spatial uncertainty quantification – *Discover Sustainability*.

An international consensus on core reproducibility items in research 2026
Rita Banzi, Mónika Varga, Yuri Andrei Gelsleichter, Constant Vinatier, David Moher, Florian Naudet
DOI: 10.1371/journal.pbio.3003726
International Delphi-based consensus defining a core set of reproducibility items for research reporting – *PLOS Biology*.

Digital Soil Mapping of Soil Organic Carbon at Eastern Slopes of Mount Kenya 2025
Gelsleichter, Y.A., Toth, A.J., Micheli, E., Mutuma, E., Wawire, A. and Csorba, A.
DOI: 10.1134/S1064229324602749
Application of Quantile Random Forest with Google Earth Engine data to map soil organic carbon on Mount Kenya

Awareness and adoption strategies for improved agricultural practices (IAPs) by smallholder farmers in the Mbeya Region, Tanzania 2025
Phenson Justine, Ádám Csorba, Caleb Melenya Ocansey, Brian Rotich, Isaiah Maket, Azaria Stephano Lameck, Mustapha Abdulkadir, Mohammed Ahmed MohammedZein, Erika Michéli, Yuri Andrei Gelsleichter
DOI: 10.1007/s43621-025-00970-y
Assess smallholder farmers' awareness and adoption of improved agricultural practices (IAPs)

Variation in Soil Organic Carbon and Total Nitrogen Stocks Across Elevation Gradients and Soil Depths in the Mount Kenya East Forest 2025
Brian Rotich, Tamás Szegi, Yuri Andrei Gelsleichter, Márta Fuchs, Caleb Melenya Ocansey, Justine Nsima Phenson, Mustapha Abdulkadir, Harison Kipkulei, Amos Wawire, Evans Mutuma, Samuel Ayodele Mesele, Erika Michéli, Ádám Csorba
DOI: 10.3390/land14061217
Soil organic carbon and total nitrogen stocks in the Mount Kenya East Forest vary significantly with elevation and soil depth, reflecting complex interactions between topography, vegetation, and soil properties – *Land*.

Spectroscopy Supported Definition and Classification of Sandy Soils in Hungary 2024
Erika Michéli, Márta Fuchs, Yuri Gelsleichter, Mohammed Zein, Ádám Csorba
DOI: 10.1007/978-3-031-50285-9_6
Analyses of taxonomic relations and distances comparing centroids calculated from legacy laboratory data and MIR spectral data of the Hungarian Soil Information and Monitoring System – Book chapter of Sandy Soils; Springer.

Enhancing Soil Mapping with Hyperspectral Subsurface Images generated from soil lab Vis-SWIR spectra tested in southern Brazil 2023
Gelsleichter, Y.A.; Costa, E.M.; Anjos, L.H.C.; Marcondes, R.A.T.
DOI: 10.1016/j.geodrs.2023.e00641
Developed Hyperspectral Soil Mapping by integrating lab-based Vis-SWIR spectra with Digital Soil Mapping to improve soil property predictions. Utilized Random Forest algorithm with k-fold cross-validation, achieving enhanced prediction accuracy for Total Carbon content in soils. Applicable in agriculture and conservation.

Past and Future Responses of Soil Water to Climate Change in Tropical and Subtropical Rainforest Systems in South America 2023
Arévalo, S.M.M. et al.
DOI: 10.3390/atmos14040755
Assessed impacts of land use and climate change on South America's tropical longleaf forests using ARIMA models for soil moisture and future climate scenarios. Identified regional trends and vulnerabilities, contributing to targeted forest preservation strategies.

Degradation of South American biomes: What to expect for the future? 2022
Delgado, R.C.; Santana, R.O.; Gelsleichter, Y.A.; Pereira, M.G.
DOI: 10.1016/j.eiar.2022.106815
Analyzed 18 years of meteorological, biophysical, and fire variables to study vegetation dynamics across South American biomes, forecasting trends until 2040. Predicted increased air temperatures will intensify evaporation, transpiration, and mega-fires, signaling a need to prevent biodiversity loss.

Mapping soil properties in a poorly-accessible area 2020
Costa, E.M.; Pinheiro, H.S.K.; Anjos, L.H.C.; Marcondes, R.A.T.; Gelsleichter, Y.A.
DOI: 10.36783/18069657rbcs20190107
Conducted soil mapping in the remote Itatiaia National Park focusing on properties like pH, carbon content, and CEC using Digital Soil Mapping tools. Employed conditioned Latin Hypercube Sampling and compared linear and nonlinear models for calibration.

Spatial Bayesian belief networks: a participatory approach for mapping environmental vulnerability at the Itatiaia National Park, Brazil 2020
Costa, E.M.; Pinheiro, H.S.K.; Anjos, L.H.C.; Gelsleichter, Y.A.; Marcondes, R.A.T.
DOI: 10.1007/s12665-020-09099-9
Implemented a Bayesian Belief Network to assess environmental vulnerability using data like soils, land use, climate, and relief. Identified areas of fragility based on environmental factors, providing actionable insights for park management.

Ongoing Publications

Submitted

- Predicting soil properties with Vis-NIR spectra in Northern Nigeria
  M. Abdulkadir, A. Csorba, A. M. Zubairu, **Y. A. Gelsleichter**

Writing phase

- Soil data treatment – Namibia case *(target: Eurasian Soil Science)*
  Yuri Gelsleichter, Marina Coetzee, Ádám Csorba, Erika Michéli
- Mapping SOC NEON sites with high spectral and spatial resolution in HPC
  Yuri Gelsleichter, Samantha Weintraub-Leff, Shashi Konduri, Jitendra Kumar, Debjani Sihi
- Hungarian spectral library – models, methods and clusters
  Ádám Csorba, **Yuri Gelsleichter**, Erika Michéli
- EJP I: Soil tillage and spectra – linear mixed models
  Yuri Gelsleichter, Tamás Szegi, Beáta
- Sewage sludge treatment I – spectral investigation, linear mixed models
  Khalid, **Yuri Gelsleichter**, Tamás Szegi, Marianna
- Influence of land use on SOC – linear mixed models
  Betim, Caleb Melenya Ocansey, **Yuri Gelsleichter**, Tamás Szegi
- Hungarian hydromorphic soils – spectroscopy, centroids and class distances
  Réka, **Yuri Gelsleichter**, Erika Michéli
- Predicting soil properties with MIR spectra in Tanzania
  Justine Phenson, **Yuri Gelsleichter**, Ádám Csorba, Erika Michéli

Data processing

- Sewage sludge treatment II – seasons and soil respiration, linear mixed models
  Khalid, **Yuri Gelsleichter**, Tamás Szegi, Marianna

- Hungarian soil classification system – automated classification
  Márta Fuchs, **Yuri Gelsleichter**

Planned

- EJP II: Soil tillage, Chapter II – linear mixed models
  Yuri Gelsleichter, Tamás Szegi
- Influence of land use on SOC, Chapter II – linear mixed models
  Betim, Caleb Melenya Ocansey, **Yuri Gelsleichter**, Tamás Szegi
- Liberian soils *(invited contribution)*
  Ádám Csorba **Yuri Gelsleichter**, Samuel Ayodele Mesele

Projects

Soils 4 Liberia
2026 - CURRENT

https://www.soils4liberia.org/

Capacity building for the Liberian soil sector, strengthening national expertise in soil survey and soil data handling.
–Participation: Local training on soil classification, data processing and spatial sampling.

Capacity building | Soil classification | Spatial sampling

AUSO – Africa Union Soil Observatory
2026 - CURRENT

https://auso.faraafrica.org/

AUSO is building a continental soil health monitoring platform, pairing a soil data hub with a dashboard to close soil data gaps and deliver site-specific information for technical, policy and infrastructure decisions across Africa.
–Participation: Technical support.

Africa | Soil monitoring | Soil data hub

OSIRIS
2023 - 2027

https://osiris4r.eu/

Open Science to Increase Reproducibility In Science (OSIRIS) aims to facilitate this shift by systematically gathering knowledge on the underlying drivers, testing effective evidence-based solutions, identifying incentives for reproducibility by stakeholders, and embedding reproducibility in research design.
–Participation: Implemented and analyzed a survey to gather the current barriers and motivations for Computational Science Reproducibility; Collaborated in the development of a checklist for science reproducibility, also with the implementation of a Delphi Survey in LimeSurvey platform, where the results were applied within the project.

Reproducibility | Open-data | Open-code

Soils 4 Africa
2024 - 2026

https://www.soils4africa-h2020.eu/

Soils4Africa aims to put in place an Open-data Soil Information System, to improve sustainable intensification of agriculture and boosting food security; by improving the quality and availability of African soil data.
–Participation: Active assistance in Portuguese-speaking countries. Gave training in Minimalist Reference Sites sampling collection for 22 participants. Protocol and field survey translation. Assistance for field surveyors of Portuguese-speaking.

Africa | Soil Information System | Open-data

Pensa Rio – Itatiaia National Park (FAPERJ)
2016 - 2020

https://pnitatiaia.com.br/

Soil survey and digital soil mapping of the plateau of Itatiaia National Park (Rio de Janeiro, Brazil). The project characterised the soils and their spatial distribution and derived an environmental vulnerability map to guide public-use restrictions and conservation planning by the park administration, funded by FAPERJ, CNPq and CAPES.
–Participation: Field survey campaigns; laboratory Vis-NIR/SWIR spectroscopy and spectroradiometric prediction of soil properties; machine learning modelling.

Digital Soil Mapping | Protected areas | Vis-NIR spectroscopy

Teaching

Soil Mapping — Oct 2022 – Current

Hungarian University of Agriculture and Life Sciences (MATE), Gödöllő, Hungary

- Conducted soil mapping projects using R programming.
- Supervised undergraduate students in data collection and analysis.
- Integrated digital soil mapping techniques into curriculum.

Introduction to R Programming Course — Feb 2022 – May 2025

Hungarian University of Agriculture and Life Sciences (MATE), Gödöllő, Hungary

- Developed and taught a 25-hour graduate-level course on R programming.
- Covered data analysis and visualization techniques.
- Enhanced students' proficiency in coding.

Soil Properties prediction with spectral data (Vis-NIR, MIR) — Nov 2024 – Current

Hungarian University of Agriculture and Life Sciences (MATE), Gödöllő, Hungary

- Data preparation and data shaping to modeling.
- Machine learning modeling to predict soil properties.

- Advanced validation strategies.

(Online) Data Analysis and Visualization in R Applied to Agronomy and Soil Sciences Oct 2020
Federal Rural University of Rio de Janeiro (UFRRJ)
- Provided a 60-hour graduate-level online course focusing on R for agronomy.
- Emphasized soil data analysis and visualization.
- Facilitated remote learning during pandemic restrictions.

Data Analysis and Visualization in R Applied to Agronomy and Soil Sciences Jun 2019
Federal Rural University of Rio de Janeiro (UFRRJ)
- Conducted a 60-hour in-person course for graduate students.
- Guided students through practical applications of R in soil sciences.
- Encouraged adoption of open-source tools for research.

Volunteering

International Soil Judging – Co-chair of Communication working group Sep 2024 – current
World wide – webpage https://soiljudgingiuss.github.io/
- Establish communication channels with National teams.

22nd World Congress of Soil Science Aug 2022
Scotland
- Assisted during the event, ensuring smooth operations and participant support.

4th International Soil Judging Contest - Stirling Aug 2022
Scotland
- Collaborated on the organization, including site preparation and profile development for participants.

Wageningen Soil Conference Aug 2019
Netherlands
- Coordinated tasks and managed team members to facilitate conference activities.

21st World Congress of Soil Science Aug 2018
Rio de Janeiro, Brazil
- Collaborated with the organizing committee, providing assistance throughout the event.

3rd International Soil Judging Contest - Seropédica Aug 2018
Rio de Janeiro, Brazil
- Assisted in organizing the event, including site setup and profile preparation for participants.